

Auditing Coherence and Partially Identifying Rankings from Sparse Pairwise Judgments

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Abstract

Sparse pairwise judgments are often converted into complete rankings even when the comparison mechanism may reflect heterogeneous criteria and the observed graph leaves substantial rank uncertainty. We develop a geometry-to-identification framework that separates three questions: compatibility with a maintained order geometry, low-dimensional parametric representation, and the rank conclusions identified by the data. The pairwise marginal Q records precedence probabilities, whereas the bistochastic marginal P records rank-position probabilities. Moon’s random-tournament theorem, Rado’s majorization theorem, and the Birkhoff–von Neumann theorem imply that every complementary Q admits a P with the same expected ranks; first-moment P – Q coupling therefore does not establish joint realizability. We audit Q using a polynomial-size triangle outer relaxation, compute an L_∞ compatibility radius and coordinate projections for pair probabilities and expected ranks, and summarize rank positions conditionally by maximum entropy. A rational seven-object certificate lies at exact L_∞ distance $1/28$ from the linear-ordering polytope while satisfying all triangle inequalities.

The CESR application contains 60 management-exam scripts and 304 single-call large-language-model comparisons. The elicited probabilities have zero radius in the triangle relaxation but remain about .05 in mean absolute error from an unpenalized Bradley–Terry (BT) fit (0.0552) and a selected two-component finite Plackett–Luce (PL) mixture (0.0544); a generative quantization null yields 0.0056. A combinatorial-Hodge analysis assigns 6.28% of squared logit-flow energy to non-gradient structure. Removing harmonic flow reduces in-sample log loss against an external human-score direction more than removing local curl, but none of the 17 Hodge-related comparisons crosses the Bonferroni reference 0.0029. At tolerances .05 and .10, mean expected-rank interval widths are 66.2% and 74.0% of the full 59-position range. On random connected exact Bradley–Terry controls matched in object and edge counts, but not graph topology, the zero-tolerance mean width is 39.47 positions, compared with 33.28 on CESR; a maximum a posteriori (MAP) tie-block method also returns blocks despite the controls having no latent ties. In a controlled finite-mixture simulation, a probability-loss improvement is detected at 200 judgments per observed edge but not at 50. These results distinguish local order compatibility, parametric representability, and rank identification.

Keywords: pairwise comparisons; partial identification; linear-ordering polytope; Birkhoff polytope; rank uncertainty; active ranking; large language model evaluation.

1 Introduction

Pairwise comparison is a natural interface for judgments that are difficult to express on an absolute scale. An evaluator may find it easier to decide which of two essays is stronger than to assign both essays calibrated numerical scores. The same observation motivates preference learning, rank aggregation, tournament design, and recent uses of large language models (LLMs) as comparative judges [Liusie et al., 2024, Shibata and Miyamura, 2025]. The statistical output, however, is often

much more definite than the input: a sparse and noisy set of comparisons is converted into a complete ordered list.

The dominant modeling move is to associate each object i with a scalar ability θ_i and to set $\Pr(i \prec j) = \sigma(\theta_i - \theta_j)$, as in the Bradley–Terry (BT) and related random-utility models [Bradley and Terry, 1952, Luce, 1959, Plackett, 1975]. This compression is valuable. It pools information, predicts unobserved pairs, and yields a ranking with only $n - 1$ effective parameters. But it also imposes a strong geometry. All pairwise probabilities must be generated by a single latent axis, and the precision of the fitted order can be mistaken for information supplied by the data.

This paper takes a different, audit-oriented view. The primitive object is a distribution μ over complete rankings. It has two useful marginal representations: a pairwise precedence matrix Q , where q_{ij} is the probability that i precedes j , and a rank-position matrix P , where P_{ik} is the probability that i occupies position k . These marginals answer different questions. Q is directly connected to comparison data and to the geometry of linear orders. P is easy to interpret but is only indirectly identified.

The distinction produces a simple but consequential modeling result. Moon’s characterization of random-tournament mean scores, Rado’s majorization theorem, and Birkhoff–von Neumann imply that any complementary pairwise matrix Q —even one not realizable by a distribution on rankings—has a permutahedral expected-rank vector and therefore admits a bistochastic P with those means. Consequently, coupling P and Q only through expected ranks cannot enforce joint realizability. This rules out an appealing but insufficient formulation and clarifies the computational architecture: impose a declared order-geometry restriction on Q , derive rank identification from Q , and treat P as a conditional reconstruction rather than an independently learned uncertainty distribution.

Our contribution is an integrated geometry-to-identification architecture for sparse pairwise assessment. It establishes why first-moment coupling of pairwise and positional marginals is insufficient, then turns an explicit order relaxation into computable compatibility radii, expected-rank projections, point reconstructions, and acquisition diagnostics. The framework provides:

- (i) a modeling consequence of the Moon–Rado characterization showing precisely what the P – Q link does and does not identify;
- (ii) a linear-programming compatibility radius and triangle-relaxed expected-rank projections under incomplete soft comparisons;
- (iii) a direct audit of the gap between triangle-compatible edge probabilities and low-dimensional scalar or finite-mixture representations;
- (iv) a maximum-entropy P reconstruction whose interpretation is conventional rather than inferential;
- (v) an acquisition heuristic that targets unresolved pair and rank intervals; and
- (vi) a controlled repetition experiment connecting learnable heterogeneity to the number of judgments collected per edge.

Five empirical findings demonstrate the diagnostic value of this separation. First, the CESR edge probabilities admit a zero-radius completion in the triangle relaxation but are not closely reproduced by either a scalar score or a selected two-component finite mixture. Second, at sensitivity tolerances chosen to match the scale of that representation gap, coordinatewise expected ranks occupy roughly two thirds to three quarters of their full amplitude, although some binary orders remain determined. Third, the Hodge decomposition separates local curl from harmonic structure and locates the largest unadjusted human-alignment contrast on the harmonic component. Fourth, a random connected exact-BT baseline matched in object and edge counts places the CESR zero-tolerance widths in context. Fifth, under a correctly specified finite-mixture truth, a predictive probability-loss improvement is detected with 200 judgments per edge but not with 50 under the reported design. An extensive propagation benchmark, reported in the appendix, complements these identification results.

2 Related work

Parametric and nonparametric comparison models. Bradley–Terry, Luce, and Plackett–Luce (PL) models derive pair probabilities or ranking probabilities from scalar utilities [Bradley and Terry, 1952, Luce, 1959, Plackett, 1975]. Efficient estimation is well established, including minorization–maximization procedures [Hunter, 2004]. Spectral alternatives such as Rank Centrality also exploit low-dimensional structure [Negahban et al., 2017]. More flexible stochastic-transitivity classes relax the exact logistic form while preserving substantive order restrictions [Shah et al., 2016, Shah and Wainwright, 2018]. Our objective is different: to distinguish a convenient point-completion model from the larger set of pairwise probabilities consistent with elementary order geometry.

Distributions over rankings and order polytopes. A distribution over permutations induces a point in the linear-ordering polytope, the convex hull of incidence vectors of total orders. Its facets and separation problems are classical subjects in polyhedral combinatorics [Fishburn, 1990, 1992, Grötschel et al., 1985, 1984]. Fishburn’s direct formulation of binary probabilities induced by rankings also records the necessity and eventual insufficiency of triangle inequalities. Random-preference and stochastic-rationalizability tests more generally ask whether observed choice probabilities admit a nonparametric distribution over preference states. For binary choice induced by linear or weak orders, Smeulders et al. [2018] develop exact membership tests based on column generation. In econometric random-utility analysis, Kitamura and Stoye [2018] construct a nonparametric statistical test, and Smeulders et al. [2021] provide column-generation tools that make it computationally usable. These works provide general membership and statistical rationalizability tests. We focus on sparse pairwise judgments and combine an L_∞ compatibility radius for a documented outer relaxation at $n = 60$ with expected-rank projections, the first-moment P – Q non-equivalence, conditional reconstruction of P , and an explicit separation of coherence, interpolation, and prediction. Mixtures of Plackett–Luce and more general random-utility models are established models for heterogeneous ranking populations; their identifiability and estimation have been studied from complete and partial rankings [Zhao et al., 2016, 2018, Liu et al., 2019, Nguyen and Zhang, 2023]. The rank-position marginal belongs to the Birkhoff polytope, whose extreme points are permutation matrices [Birkhoff, 1946]. Landau characterized deterministic tournament score sequences, Moon extended the result to mean scores of random tournaments, and Rado characterized permutahedra by majorization [Landau, 1953, Moon, 1963, Rado, 1952]. Recent geometric treatments make the projection from tournament probabilities to the permutahedron explicit [Aldous and Kolesnik, 2022, Kolesnik and Sanchez, 2024]. We use these established results to expose a modeling non-equivalence: agreement of P and Q on expected ranks is necessary but not sufficient for a common latent distribution.

Partial identification and active ranking. Partial identification replaces an unjustifiably precise estimator by the set of parameter values compatible with maintained assumptions and observations [Manski, 2003]. Pairwise-ranking work has also studied which comparisons should be requested to recover a total order or a top- K set efficiently [Jamieson and Nowak, 2011, Mohajer et al., 2017, Heckel et al., 2019]. Our acquisition score is an interpretable heuristic for shrinking a polyhedral identified set; optimal design under minimax criteria is a separate problem.

Our use of the term partial identification is closest in spirit to revealed preference analyses that recover bounds on rankings from incomplete stochastic choice information [Sher et al., 2011]. It also differs from *rankability* indices, which quantify how closely data resemble an ideal dominance hierarchy or how much a ranking problem admits a stable linear order [Anderson et al., 2019, 2021]. Here the target is a coordinatewise projection of a maintained polyhedral model: each reported interval answers which expected ranks remain compatible with the observed probabilities and a declared tolerance.

Crippa and Fedchenko analyze rankings under incomplete pairwise interactions. They derive sharp identification regions for latent merit using either parametric Bradley–Terry–Luce structure or a nonparametric stochastic-transitivity class, together with membership tests [Crippa and Fedchenko, 2024]. Our target differs: we bound expected ranks induced by pairwise marginals of arbitrary random rankings under a documented polyhedral outer relaxation, and separate this maintained set from interpolation, predictive completion, and conditional reconstruction of P .

Recent work on ranking with confidence instead starts from a generative noisy comparison model and attaches confidence measures to its estimates [Valdeira and Soares, 2025]. Our intervals answer a different question: they are deterministic projections of a maintained robust set and are not claimed to be calibrated posterior or frequentist confidence intervals. Neuhoﬀ and Benjamini instead construct hierarchical rank intervals with statistical guarantees for multi-task leaderboards [Neuhoﬀ and Benjamini, 2026]; this usefully contrasts with our deterministic coordinate projections under explicitly maintained pairwise constraints.

Morel-Balbi and Kirkley instead use a Bayesian model and maximum-a-posteriori (MAP) agglomeration to return partial rankings with ties when sparse noisy comparisons do not support a strict order [Morel-Balbi and Kirkley, 2025]. That objective is closely related but distinct: our output is a collection of deterministic coordinate projections of a robust polyhedral set, not a posterior or MAP partial order. The Mixture Linear Ordering Problem addresses heterogeneous populations by recovering several latent linear orders and their weights through mixed-integer optimization [Aledo et al., 2026]. Our finite Plackett–Luce mixture serves as a separate predictive benchmark for this broader class of heterogeneous models.

LLMs as comparative judges. Pairwise prompting can improve agreement and reduce calibration demands in automatic evaluation [Liusie et al., 2024]; it has recently been adapted to automated essay scoring [Shibata and Miyamura, 2025]. At the same time, LLM judgments may exhibit order and position effects [Shi et al., 2025]. This makes coherence auditing especially relevant. A model can produce stable-looking scores even when its local comparisons reveal plural criteria, cycles, prompt-position bias, or insufficient coverage.

3 Pairwise and rank-position marginals

Let $[n] = \{1, \dots, n\}$ denote the objects and let $\mathfrak{S}_n$ be the set of permutations. We write $\pi(i) \in [n]$ for the rank position of object i , with position 1 best. A latent distribution $\mu \in \Delta(\mathfrak{S}_n)$ represents uncertainty or heterogeneity over complete rankings; the analysis operates through its marginals rather than its $n!$ masses.

Definition 1 (Pairwise precedence marginal). *For $i \neq j$, define*

$$q_{ij} = \Pr_{\pi \sim \mu} \{\pi(i) < \pi(j)\}. \quad (1)$$

Then $q_{ij} + q_{ji} = 1$ and $q_{ii} = 0$ by convention.

Definition 2 (Rank-position marginal). *For $i, k \in [n]$, define*

$$P_{ik} = \Pr_{\pi \sim \mu} \{\pi(i) = k\}. \quad (2)$$

The matrix P is bistochastic: $P\mathbf{1} = \mathbf{1}$, $P^\top \mathbf{1} = \mathbf{1}$, and $P \geq 0$.

Both marginals determine the expected rank

$$r_i := \mathbb{E}_\mu[\pi(i)] = \sum_{k=1}^n k P_{ik} = 1 + \sum_{j \neq i} q_{ji}. \quad (3)$$

The second equality follows because the rank of i is one plus the number of objects preceding it.

The exact feasible sets are convex hulls. Let $X_{ij}^\pi = \mathbb{1}\{\pi(i) < \pi(j)\}$ and let $M_{ik}^\pi = \mathbb{1}\{\pi(i) = k\}$. Then

$$\mathcal{L}_n = \text{conv}\{X^\pi : \pi \in \mathfrak{S}_n\}, \quad (4)$$

$$\mathcal{B}_n = \text{conv}\{M^\pi : \pi \in \mathfrak{S}_n\}. \quad (5)$$

The first is the linear-ordering polytope and the second the Birkhoff polytope. If P and Q arise from the same μ , they belong to their respective polytopes and satisfy Equation (3). The converse based only on these marginal restrictions is false.

4 Geometry of pairwise and position marginals

4.1 Expected ranks of a complementary pairwise matrix

Consider the broad class

$$\mathcal{C}_n = \{Q \in [0, 1]^{n \times n} : q_{ii} = 0, q_{ij} + q_{ji} = 1 \text{ for } i \neq j\}. \quad (6)$$

For $Q \in \mathcal{C}_n$, define $r_i(Q) = 1 + \sum_{j \neq i} q_{ji}$.

Proposition 3 (Moon–Rado consequence for mean ranks). *For every $Q \in \mathcal{C}_n$, the vector $r(Q)$ belongs to the permutahedron*

$$\Pi_n = \text{conv}\{(\pi(1), \dots, \pi(n)) : \pi \in \mathfrak{S}_n\}. \quad (7)$$

Equivalently, there exists a bistochastic matrix $P \in \mathcal{B}_n$ such that $P(1, \dots, n)^\top = r(Q)$.

Proof. First, complementarity gives

$$\sum_{i=1}^n r_i(Q) = n + \sum_i \sum_{j \neq i} q_{ji} = n + \binom{n}{2} = \frac{n(n+1)}{2}. \quad (8)$$

For any subset $S \subset [n]$ of size k ,

$$\sum_{i \in S} r_i(Q) = k + \sum_{i \in S} \sum_{j \neq i} q_{ji} \quad (9)$$

$$\geq k + \sum_{\{i,j\} \subset S} (q_{ij} + q_{ji}) = k + \binom{k}{2} = \frac{k(k+1)}{2}. \quad (10)$$

These total-sum and subset-sum inequalities are the ascending-rank form of Moon’s mean-score conditions and, by Rado’s majorization theorem, characterize the convex hull of permutations of $(1, \dots, n)$. Hence $r(Q) \in \Pi_n$. The equivalence follows from the Birkhoff–von Neumann theorem: $P \in \mathcal{B}_n$ is a convex combination of permutation matrices, and multiplying by $(1, \dots, n)^\top$ yields a convex combination of rank permutations. $\square$

Corollary 4 (First-moment redundancy of P). *If a formulation imposes only $Q \in \mathcal{C}_n$, $P \in \mathcal{B}_n$, and the expected-rank link Equation (3), then P adds no restriction to Q .*

Thus a bistochastic P neither restricts nor repairs Q under first-moment coupling, and it can be reconstructed after Q has been estimated or bounded. The existence of such a P does not imply that P and Q are marginals of a common μ ; joint realizability requires richer constraints.

4.2 A polynomial-size outer relaxation

Exact membership in $\mathcal{L}_n$ is computationally difficult. A basic necessary condition comes from every triple of distinct objects. With a fixed orientation $i < j < k$, any total order satisfies

$$0 \leq q_{ij} + q_{jk} - q_{ik} \leq 1. \quad (11)$$

Together with box and complementarity constraints, these inequalities define the triangle relaxation $\mathcal{T}_n$. We have

$$\mathcal{L}_n \subseteq \mathcal{T}_n \subseteq \mathcal{C}_n. \quad (12)$$

The first inclusion is an equality for $n \leq 5$ and is strict for $n \geq 6$; Möbius-ladder and other higher-order facets are then needed to describe the linear-ordering polytope [Fishburn, 1992, Grötschel et al., 1985, Fiorini, 2006]. Appendix A.3 gives our explicit rational $n = 7$ witness with zero triangle violation and L_∞ distance $1/28$ from the exact polytope. Thus every interval computed below is conditional on the triangle relaxation unless cutting planes are added. As an outer relaxation, it is conservative for rank bounds: it may be too wide, but it will not exclude a truly realizable Q merely because higher-order facets were omitted.

The relaxation has $m = n(n-1)/2$ free pair variables and $2\binom{n}{3}$ triangle inequalities. At $n = 60$, this is 1,770 variables and 68,440 oriented inequalities, readily represented as sparse matrices. The cubic constraint count, rather than the number of pair variables, is the practical scaling limit of the present implementation.

5 Triangle compatibility auditing and partial identification

Let $E \subset \{(i, j) : i < j\}$ be the observed comparison graph. Each observed edge has a soft judgment $\hat{p}_{ij} \in [0, 1]$, interpreted as an estimate of the probability that i precedes j . Hard outcomes are the special case $\hat{p}_{ij} \in \{0, 1\}$.

5.1 The triangle compatibility radius

Definition 5 (Triangle compatibility radius). *The minimal uniform correction required to make the observations feasible is*

$$\rho_\infty(\hat{p}; E) = \min_{Q \in \mathcal{T}_n} \max_{(i,j) \in E} |q_{ij} - \hat{p}_{ij}|. \quad (13)$$

Introducing a scalar t and inequalities $-t \leq q_{ij} - \hat{p}_{ij} \leq t$ makes Equation (13) a linear program. A zero radius establishes that all observed probabilities extend to a point in $\mathcal{T}_n$. Membership in $\mathcal{L}_n$ and representation by a scalar random-utility model are separate, stronger properties. A positive radius is the smallest worst-edge modification required under the maintained relaxation.

The radius complements direct triangle counts. Fully observed triples permit direct residual inspection, but sparse graphs may contain few or no such triples. The optimization also uses latent unobserved edges and can therefore detect whether the entire observed pattern admits a triangle-relaxed completion.

5.2 Identified sets and rank intervals

Let $\varepsilon_{ij} \geq 0$ encode edge-specific uncertainty, prompt instability, or a common robustness tolerance. Define

$$\mathcal{Q}_\varepsilon(E, \hat{p}) = \{Q \in \mathcal{T}_n : |q_{ij} - \hat{p}_{ij}| \leq \varepsilon_{ij}, (i, j) \in E\}. \quad (14)$$

When $\mathcal{Q}_\varepsilon$ is nonempty, the sharp bounds under the triangle-relaxed model are

$$\underline{r}_i = \min_{Q \in \mathcal{Q}_\varepsilon} r_i(Q), \quad \bar{r}_i = \max_{Q \in \mathcal{Q}_\varepsilon} r_i(Q). \quad (15)$$

Both are linear programs because $r_i(Q)$ is affine. Pairwise bounds $\underline{q}_{ij}$ and $\bar{q}_{ij}$ are obtained analogously. The collection of coordinate intervals is a set of one-dimensional projections. Its Cartesian-product box is generally not jointly attainable: different coordinates can achieve their extrema at different matrices $Q \in \mathcal{Q}_\varepsilon$. We therefore never interpret simultaneous endpoint choices as one admissible ranking vector.

Proposition 6 (Coverage under feasibility). *Suppose the data-generating pairwise matrix $Q^* \in \mathcal{L}_n$ satisfies $|q_{ij}^* - \hat{p}_{ij}| \leq \varepsilon_{ij}$ for every observed edge. Then*

$$r_i(Q^*) \in [\underline{r}_i, \bar{r}_i] \quad \text{for every } i. \quad (16)$$

Proof. Since $\mathcal{L}_n \subseteq \mathcal{T}_n$ and the edge inequalities hold, $Q^* \in \mathcal{Q}_\varepsilon$. The result follows immediately from minimization and maximization over a set containing Q^* . $\square$

This proposition gives deterministic coverage under the stated edge bands. For statistical coverage, the tolerances ε_{ij} must be calibrated from repeated comparisons, judge variability, or other sampling information. Zero-tolerance analyses instead treat the elicited probabilities as fixed inputs.

5.3 Maximum-entropy summary of rank-position marginals

Given a point estimate $\hat{Q}$ and its expected ranks $\hat{r} = r(\hat{Q})$, we reconstruct a rank-position matrix by maximum entropy:

$$\hat{P} = \arg \max_{P \geq 0} - \sum_{i,k} P_{ik} \log P_{ik} \quad \text{s.t.} \quad P\mathbf{1} = \mathbf{1}, \quad P^\top \mathbf{1} = \mathbf{1}, \quad P(1, \dots, n)^\top = \hat{r}. \quad (17)$$

Proposition 7 (Existence and uniqueness). *The feasible set is nonempty for every complementary $\hat{Q}$. The entropy objective has a unique maximizer, including when the feasible set lies on a proper face of $\mathcal{B}_n$.*

Proof. Feasibility follows from [Theorem 3](#); existence follows from compactness of $\mathcal{B}_n$ and continuity with the convention $0 \log 0 = 0$. Shannon entropy is strictly concave on every convex face after coordinates that are identically zero are removed, so two distinct maximizers would make their midpoint strictly better. $\square$

In the relative interior we solve the exponential-family dual; near a boundary face the implementation uses the primal conic program. For a row P_i , we report the entropy-based effective support $\exp\{-\sum_k P_{ik} \log P_{ik}\}$ and treat entries below 10^{-8} as numerical zero. The result is a diffuse maximum-entropy matrix compatible with the mean ranks; posterior position probabilities require an additional probabilistic model.

6 Constrained interpolation with Bradley–Terry completion

The identified set is the primary uncertainty object, but applications may also need one completed matrix for interpolation and display. We use Bradley–Terry only as a regularizing reference, not as a prior distribution or maintained truth.

Let $s_i \in \mathbb{R}$ be fitted Bradley–Terry scores, with larger scores indicating better objects, and

$$q_{ij}^{\text{BT}} = \text{sigmoid}(s_i - s_j) \quad (18)$$

under the convention that smaller rank is better and $q_{ij} = \Pr(i \text{ precedes } j)$. Scores are estimated on the observed graph by soft-label logistic loss with a centering constraint.

For the m free variables $q = (q_{ij})_{i < j}$, define the hard-constrained estimator

$$\begin{aligned} \hat{q} = \arg \min_q \quad & \frac{1}{\sum_{(i,j) \in E} a_{ij}} \sum_{(i,j) \in E} a_{ij} \left[-\hat{p}_{ij} \log q_{ij} - (1 - \hat{p}_{ij}) \log(1 - q_{ij}) \right] \\ & + \frac{\lambda}{m} \sum_{i < j} \text{KL}(\text{Bern}(q_{ij}) \parallel \text{Bern}(q_{ij}^{\text{BT}})) \\ \text{subject to} \quad & q \in \mathcal{T}_n, \quad |q_{ij} - \hat{p}_{ij}| \leq \varepsilon_{ij} \quad ((i, j) \in E), \quad \delta \leq q_{ij} \leq 1 - \delta. \end{aligned} \quad (19)$$

Here a_{ij} is an optional judgment weight, $\lambda > 0$, and the box offset is $\delta = 10^{-5}$. The first term fits observed soft labels and the second stabilizes unobserved pairs. The point estimator belongs to the same $\mathcal{Q}_\varepsilon$ used for the reported robust intervals. For an exact binary observation, feasibility with the interior box requires $\varepsilon_{ij} \geq \delta$; alternatively, the boundary can be handled directly by setting $\delta = 0$ and using the limiting objective.

Proposition 8 (Convexity of the completion objective). *For fixed $Q^{\text{BT}} \in (0, 1)^m$, problem Equation (19) is a convex program. If its constraint set is nonempty and $\lambda > 0$, it has a unique minimizer.*

Proof. The Bernoulli cross-entropy is convex in its prediction argument and $\text{KL}(\text{Bern}(q) \parallel \text{Bern}(p))$ is strictly convex in q for fixed $p \in (0, 1)$. All triangle, edge-band, and box constraints are affine. $\square$

We solve the exponential-cone formulation with CLARABEL and use SCS as a fallback; primal residuals are checked after every solve. The reported reference strength is $\lambda = 10$; a four-point sensitivity analysis is reported below, while ε is fixed independently of predictive performance.

Because the KL term is separable and anchored at Q^{BT} , an unobserved coordinate remains exactly at its Bradley–Terry value whenever the order constraints involving that coordinate are inactive. The estimator is therefore a *constrained interpolation with Bradley–Terry completion*, not by itself a propagative prediction model.

For a held-out predictive-propagation benchmark, we fit established finite Plackett–Luce mixtures with one to four components and compare them with BT under common inner edge splits, penalty grids, and 12-start final refits. The complete objective, initialization, selection, and stability protocol is given in Appendix D. The geometric quantities—the triangle compatibility radius, identified set, rank projections, acquisition scores, and P reconstruction—are computed independently of this benchmark.

7 Interval-driven comparison acquisition

When another comparison can be requested, the identified set indicates where information is missing. For each unobserved pair, solve two LPs for $[\underline{q}_{ij}, \bar{q}_{ij}]$ and use the current rank widths $d_i = \bar{r}_i - \underline{r}_i$. We score the pair by

$$A(i, j) = (\bar{q}_{ij} - \underline{q}_{ij})(d_i + d_j). \quad (20)$$

The first factor favors unresolved relationships; the second favors relationships involving poorly identified objects. A batch consists of the highest-scoring unobserved pairs, after which the identified set is recomputed.

The criterion is myopic and can allocate early comparisons to apparently influential objects before the global geometry has been mapped. The experiments accordingly combine its later contraction gains with an initial exploration phase; Equation (20) is an operational heuristic rather than an optimal experimental-design policy.

8 Empirical design

The empirical analysis addresses five questions: whether the elicited CESR probabilities are compatible with the maintained order geometry; how closely low-dimensional parametric models reproduce them; how residual curl and harmonic structure align with an external scalar order; which binary orders and expected ranks remain identified under declared sensitivity bands; and how predictive recovery changes across four levels of replication in a controlled finite-mixture simulation.

Metrics and intervals. For a true or elicited edge probability p_e and a prediction $\hat{p}_e$, mean squared probability error is

$$\text{MSPE} = |H|^{-1} \sum_{e \in H} (\hat{p}_e - p_e)^2.$$

This is squared error against a probability target, not the outcome-level Brier score that would require binary realizations. We also report soft-target cross-entropy and directional accuracy away from exact ties. Numerical contrast intervals use 5,000 paired bootstrap resamples of whole trials. Because the 30 CESR edge splits overlap, these intervals summarize variation across the specified splits rather than repeated-sample uncertainty.

CESR comparisons. The application contains 60 pseudonymized management-exam scripts and 304 distinct unordered comparisons, 17.2% of the 1,770 possible pairs. Every pair was judged once against the full question and correction guide. The acquisition program logged the mutable alias `gpt-5.5`, used `/v1/chat/completions`, omitted `temperature` and `reasoning_effort`, and did not counterbalance A/B order. Elicited probabilities were clipped to $[.01, .99]$. These values are model declarations, not empirical frequencies or calibrated probabilities. Human grades enter only an external association analysis and never fitting, tuning, or edge selection.

Quantization audit. Because a judge may verbalize probabilities on a coarse grid, a smooth model can have nonzero approximation error even when its latent mechanism is exactly scalar. We record the empirical support of the 304 declarations and construct a generative quantization null. An unpenalized BT model is fitted to the observed edges, and its predictions are rounded to the nearest value on either the empirical support or a regular grid with step .01, .05, or .10. Refitting BT to these generated targets isolates the approximation error induced by quantization from the residual structure of the CESR declarations.

Combinatorial-Hodge audit. Let $B_1 \in \mathbb{R}^{|E| \times n}$ be the signed node-edge incidence matrix of the observed graph, oriented by label, and let B_2 contain the signed boundaries of its filled observed triangles. For the edge flow $y_{ij} = \text{logit}(\hat{p}_{ij})$, combinatorial Hodge theory gives the orthogonal decomposition

$$y = B_1 s + B_2 c + h, \quad B_1^\top h = 0, \quad B_2^\top h = 0. \quad (21)$$

The first term is the best least-squares scalar potential on logits, the second is local curl supported by filled triangles, and h is the harmonic component carried by graph cycles not filled by observed triangles [Jiang et al., 2011]. We report squared-norm energy shares. The decomposition characterizes the topology of residual variation; its behavioral origin requires separate evidence.

Two Monte-Carlo nulls use 2,000 trials. The quantization null stochastically rounds each fitted BT probability to its two neighboring support values after canonical edge orientation, with probabilities chosen to preserve the fitted value in expectation whenever it lies inside that support and boundary clipping otherwise; it preserves the graph and declaration granularity. A second

null permutes the observed logit residuals over edges, then subjects each permutation to the full $n = 60$ triangle-relaxation LP. A draw is retained only when its L_∞ compatibility radius is zero up to 10^{-9} . The reference distribution therefore lives inside the same maintained geometry as the data rather than being inflated by incoherent perturbations.

Raw component energies are not directly comparable because the local-curl and harmonic subspaces have different dimensions. We use the concentration statistic

$$T = \frac{\|h\|_2^2/d_h}{\|B_2c\|_2^2/d_c}, \quad (22)$$

where d_h and d_c are the harmonic and curl dimensions. Under isotropic allocation of residual energy, T is near one; $T > 1$ means that energy is denser on global unfilled cycles than on filled-triangle boundaries. The Monte-Carlo upper-tail p -value compares observed T with 2,000 accepted permutations and includes the observation in numerator and denominator. The Monte-Carlo assessment of T is interpreted jointly with the other component diagnostics below.

The Hodge split depends on the sampled comparison graph. We assess this design sensitivity through 1,000 connected edge-deletion subsamples at each of 80, 90, and 95 percent retention. Deletion, rather than with-replacement edge bootstrap, preserves a simple graph and avoids duplicate-edge conventions.

Confidence-weighted and external-alignment audits. The judge recorded a confidence $w_e \in [.62, .97]$ on every observed edge. We first ask whether the unweighted harmonic energy h_e^2 is associated with confidence, using Spearman correlation and the ratio of mean energy below the empirical lower-quartile threshold to that above the upper-quartile threshold. We then repeat the decomposition in the weighted edge inner product $\langle x, y \rangle_W = x^\top W y$, $W = \text{diag}(w_e)$. In transformed coordinates $z = W^{1/2}y$, exact flows span $W^{1/2}B_1$ and coexact flows span $W^{-1/2}B_2$; these spaces remain orthogonal because $B_1^\top B_2 = 0$. This defines the weighted density ratio T_W . Because confidence is produced by the same LLM, it serves as an internal diagnostic weight rather than an external validation measure.

Human scores supply an external scalar order on 287 of the 304 observed edges; 17 tied-score edges are excluded. We compare the full flow y with three ablations: the deharmonized flow $y - h$, the decurled flow $y - B_2c$, and the gradient-only flow B_1s . Binary log loss and Brier loss measure probability-scale alignment with the human direction; directional accuracy measures the induced binary decision. The lead structural contrast is the improvement after removing h minus the improvement after removing B_2c . We also compare mean h_e^2 on human-LLM directional mismatches with its mean on agreements. These component contrasts form part of the 17-comparison Hodge family; their raw Monte-Carlo tail probabilities are evaluated against the familywise reference below.

Human score differences form a gradient, so Euclidean projection necessarily improves when either non-gradient component is removed. Evidence therefore rests on the nonlinear loss contrasts—especially harmonic versus curl—evaluated against the admissible permutation reference. Every confidence and human statistic is compared with the same 2,000 residual permutations accepted as triangle-admissible at the 10^{-9} feasibility tolerance.

For multiplicity accounting, the Hodge interpretation family contains 17 Monte-Carlo tail comparisons: the unweighted density ratio T , three confidence or weighted diagnostics, and thirteen external human-alignment diagnostics. Tables retain raw tail probabilities to show the declared null distributions. As a simple familywise reference, a Bonferroni threshold at level .05 is 0.0029; none of the comparisons crosses it, and the smallest raw probability 0.0040 corresponds to adjusted value 0.068. The component-specific log-loss contrast has the smallest raw tail probability in this family.

Table 1: Full-data CESR representation audit. Errors are absolute deviations from the 304 elicited edge probabilities. The selected BT and finite-PL rows use the same fixed inner-selection protocol; the unpenalized BT fit is a connected-graph sensitivity.

Model	Free par.	Par./edge	Mean error	Max. error
Selected BT	59	0.194	0.0584	0.2856
Unpenalized BT	59	0.194	0.0552	0.2874
Selected PL mixture ($K = 2$)	119	0.391	0.0544	0.2899

Assumption-ladder comparison. We compare the CESR conclusions with the two closest neighboring outputs. For Crippa and Fedchenko’s population strong-stochastic-transitivity (SST) model, the declared soft probability on an observed edge is treated as the interaction probability. Their Theorem 2, condition (3.1), requires the signs relative to one half to agree with one weak ordering; a directed majority cycle therefore certifies that the sharp SST ranking set is empty before its higher-order probability inequalities are considered [Crippa and Fedchenko, 2024]. We count such cycles and the smallest common L_∞ band that could remove this directional obstruction. Their repeated-outcome inference procedure requires replicated binary outcomes, whereas CESR contains one probability declaration per edge.

Morel-Balbi and Kirkley’s method instead takes binary match outcomes and returns one MAP partition into ordered tie blocks under a BT likelihood and a nonparametric partition prior [Morel-Balbi and Kirkley, 2025]. We run the authors’ exact greedy algorithm at public commit `ebbce35` on the 296 categorical non-tie CESR judgments; the eight declared ties are omitted because the input model excludes tied matches. An object in a block occupying positions a to b is assigned within-block position span $b - a$, and pairs in different ordered blocks are counted as resolved. We repeat the same conversion on 30 random connected exact-BT graphs matched in object and edge counts, using the oracle direction of each observed probability. On these controls we also retain the implementation’s log posterior odds of the selected partition against its BT state. The resulting spans summarize one deterministic MAP partition. Each table row therefore states its distinct estimand and assumptions: expected-rank projections under arbitrary random rankings, latent-merit rankings under SST, or a MAP deterministic ranking with ties. Posterior uncertainty over partitions is outside this comparison. The neighboring-method rows use no tail-probability calculation and are not part of the 17-comparison Hodge family.

9 Results

9.1 Local order coherence without low-dimensional representation

The 304 CESR probabilities have triangle-relaxed compatibility radius $\rho_\infty = 0$. None of the 299 fully observed triples violates Equation (11). Thus the observations admit a zero-radius completion in $\mathcal{T}_n$. The condition is weakly selective on this sparse graph: 80.1% of residual permutations also have radius zero, and the sparse-graph contamination experiment detects none of 20 departures at amplitude .05. At 17.2% edge density, $\rho_\infty = 0$ therefore defines the maintained local geometry while leaving stronger global coherence properties unresolved.

Scalar representability is considerably more restrictive. Table 1 compares the selected BT and finite-mixture fits on the full data. The selected BT fit misses the elicited probabilities by 0.0584 on average and 0.2856 at worst. Removing its penalty changes the values only to 0.0552 and 0.2874. The selected two-component finite-PL mixture, with 119 free parameters, has mean and maximum errors 0.0544 and 0.2899. It is coherent by construction, yet remains far from many elicited edges.

The nearly equal BT and finite-mixture errors identify a representational gap relative to the

Table 2: CESR pair and expected-rank identification under the triangle relaxation. Binary-order projections are computed at tolerance .05; expected-rank width sensitivities are shown at all four tolerances. The projections do not form a jointly attainable box.

ε	Determined pairs	Determined missing pairs	Mean rank width	Minimum	Maximum
0	–	–	33.28	–	40.36
0.02	–	–	35.74	–	42.58
0.05	449/1770 (25.4%)	154/1466 (10.5%)	39.05	25.49	45.47
0.1	–	–	43.64	–	49.06

Table 3: Finite three-component PL positive control as the number of Bernoulli judgments per observed edge increases. Entries are finite-PL-mixture minus BT contrasts on unobserved true pair probabilities with paired bootstrap intervals over 30 trials.

Judgments per edge	MSPE			Cross-entropy			Directional accuracy	
	Mean	Low	High	Mean	Low	High	Mean	95% interval
5	+0.00383	+0.00038	+0.00813	+0.01785	+0.00119	+0.03983	+0.0111	[-0.0074, +0.0370]
20	+0.00034	-0.00094	+0.00199	+0.00125	-0.00192	+0.00510	+0.0111	[-0.0037, +0.0259]
50	-0.00141	-0.00308	+0.00012	-0.00266	-0.00597	+0.00081	+0.0296	[+0.0037, +0.0593]
200	-0.00322	-0.00428	-0.00220	-0.00672	-0.00887	-0.00461	+0.0741	[+0.0333, +0.1185]
∞ (oracle)	-0.00613	-0.00738	-0.00494	-0.01278	-0.01537	-0.01028	+0.1519	[+0.1111, +0.1926]

maintained triangle relaxation. The observations have $\rho_\infty = 0$, but neither tested low-dimensional family reproduces them to better than roughly five probability points on average. Local triangle compatibility therefore leaves substantially more freedom than a scalar score or this selected small mixture. This statement concerns the observed edge probabilities, not graph parsimony or held-out prediction.

9.2 Quantization is an order of magnitude too small

The declarations are not concentrated on a five- or ten-point verbal grid. They contain 62 distinct values, with minimum positive step 0.01; only 12.8% and 3.3% are multiples of .05 and .10, respectively. The largest repeated value occurs 20 times. The empirical histogram and generative nulls are shown in Figure 1 and Table 5.

On the actual support, the refitted quantized-BT null has mean absolute error 0.0056, compared with 0.0552 on CESR: the observed discrepancy is 9.9 times larger. Even the counterfactual .10 grid yields mean error only .0235. Under the reported quantization nulls, grid rounding accounts for only a small part of the measured approximation error and does not reproduce the five-point BT discrepancy on the observed support.

9.3 Harmonic concentration beyond local coherence

The observed graph has 304 edges and 299 filled triangles. Its edge space decomposes into a 59-dimensional gradient space, a 172-dimensional local-curl space, and a 73-dimensional harmonic space. As reported in Table 6 and Figure 2, the CESR logit flow assigns 6.28% of its squared energy to the non-gradient component: 3.88% is local curl and 2.40% is harmonic.

Under the stochastic quantized-BT null, the mean non-gradient share is only 0.40%; none of 2,000 trials reaches the observed share ($p = 1/2001$). Under this null, quantization noise on the observed support is too small to reproduce either the probability error or the observed non-gradient energy.

Component dimensions make raw energy comparisons misleading. Curl energy is 36.41 over

Table 4: Random connected exact-BT baseline for expected-rank widths: $n = 60$, 304 uniformly sampled connected edges, and zero tolerance.

Trials	Mean width	2.5%	97.5%	Fraction of full 59-position range
30	39.47	39.05	39.81	66.9%

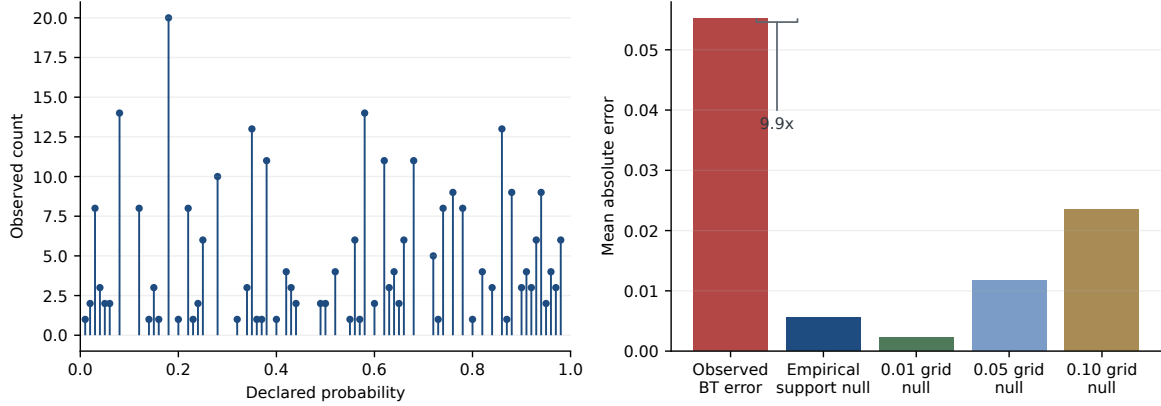


Figure 1: Quantization audit. Left: the 62-value empirical support of the 304 declared probabilities. Right: observed BT approximation error versus generative quantized-BT nulls after refitting. The empirical-support null is approximately ten times smaller than the observed discrepancy.

172 dimensions, or 0.212 per dimension; harmonic energy is 22.47 over 73 dimensions, or 0.308. The harmonic-to-curl density ratio is therefore $T = 1.454$: global-cycle energy is 45.4% denser per available direction than filled-triangle curl energy.

The residual-permutation null is conditioned on the maintained geometry. Only permutations accepted as triangle-admissible at the 10^{-9} feasibility tolerance are retained; 80.1% of evaluated proposals satisfy that condition. Among 2,000 accepted permutations, mean T is 0.972 with a 95% range $[0.642, 1.390]$. The observed ratio lies in the upper tail: 31 null draws reach or exceed it, and the finite-sample correction gives numerator 32 and raw upper-tail $p = 0.0160$. The observed pattern combines zero-radius triangle compatibility with a non-scalar component whose energy is denser per dimension on global, unfilled cycles than on local triangle boundaries.

Triangle inequalities act on probabilities and do not force logit-flow curl to vanish. With one call and one prompt orientation per edge, the available data also cannot distinguish stable heterogeneous criteria, order effects, and model instability as behavioral sources of the observed flow.

Mean concentration changes little under the reported deletion analysis, but it remains design-dependent. Mean T is 1.431, 1.421, and 1.402 when 95, 90, and 80 percent of edges are retained. Harmonic energy density rises monotonically from 0.285 through 0.295 and 0.301 to 0.308 on the full graph, while curl density remains nearly flat. Even at 80 percent, the 95% deletion range for T is $[1.077, 1.820]$. These deletions measure sensitivity within the observed design; alternative acquisition plans and the 82.8% of pairs never judged remain outside this calculation.

Table 5: Quantization audit. Each null is generated by rounding the unpenalized BT fit, refitting BT to the rounded targets, and measuring error against those targets. This generative refit quantifies the approximation error induced by support rounding under the stated null.

Data or null	Support size	Mean error	Maximum error	Observed / null
CESR versus unpenalized BT	62	0.0552	0.2874	1.0
BT null, empirical support	62	0.0056	0.0246	9.9
BT null, 0.01 grid	101	0.0023	0.0078	24.0
BT null, 0.05 grid	21	0.0118	0.0331	4.7
BT null, 0.10 grid	11	0.0235	0.0698	2.3

Table 6: Combinatorial-Hodge decomposition of the CESR logit edge flow and quantized-BT null comparison. Energy shares are squared Euclidean norms; the null row is the mean over 2,000 trials.

Flow or null	Gradient	Local curl	Harmonic	Non-gradient
CESR logit flow	93.72%	3.88%	2.40%	6.28%
Stochastic quantized-BT null, mean	99.60%	0.34%	0.06%	0.40%

Table 7: Harmonic concentration and design stability. Energy density is squared logit-flow energy divided by subspace dimension. The null retains 2,000 residual permutations accepted as triangle-admissible at the 10^{-9} feasibility tolerance. Edge-deletion intervals are empirical 2.5–97.5 percentiles over 1,000 connected subsamples at each retention rate.

Flow or design	Curl density	Harmonic density	Ratio T	95% range
CESR, full graph	0.212	0.308	1.454	—
Admissible residual-permutation null	0.209	0.200	0.972	[0.642, 1.390]
Retain 95% of edges	0.210	0.301	1.431	[1.245, 1.660]
Retain 90% of edges	0.209	0.295	1.421	[1.178, 1.709]
Retain 80% of edges	0.205	0.285	1.402	[1.077, 1.820]

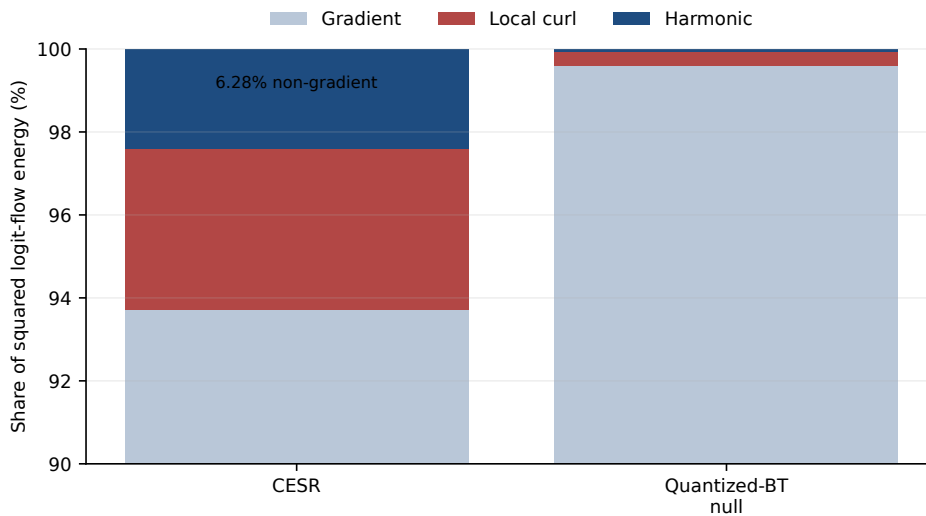


Figure 2: Combinatorial-Hodge energy decomposition of the CESR logit edge flow and mean under the quantized-BT null. The vertical axis is truncated to show the non-gradient components; all bars sum to 100%.

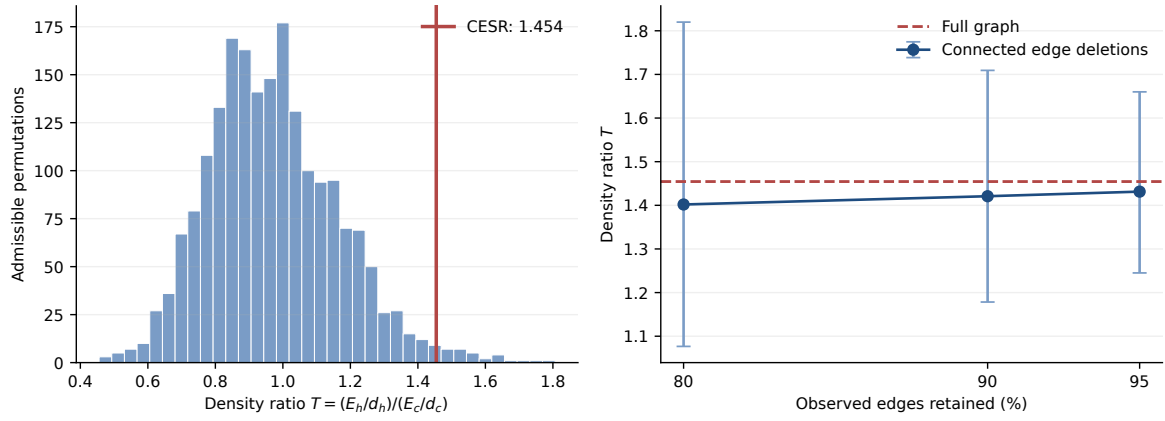


Figure 3: Harmonic concentration under the maintained geometry. Left: null distribution of the energy-density ratio T over 2,000 residual permutations accepted as triangle-admissible at the 10^{-9} feasibility tolerance. Right: mean T and empirical 95% ranges under connected edge deletion; the dashed line is the full graph.

9.4 External calibration localizes unusual misalignment to global cycles

Confidence weighting leaves the density ratio nearly unchanged. The weighted density ratio is $T_W = 1.430$, compared with null mean 0.943 and 95% range [0.621, 1.351]; the upper-tail probability remains $p = 0.0160$. The observed concentration therefore persists when edges are weighted by the model’s declared confidence.

The edgewise confidence diagnostic lies within its reference distribution. Harmonic energy has Spearman association -0.095 with declared confidence, and its mean is 1.646 times as large in the low-confidence group as in the high-confidence group. Yet admissible permutations produce mean association -0.128 and mean ratio 1.613; the observed ratio has $p = 0.420$. Although low-confidence edges carry more harmonic energy, the magnitude is typical under the fixed-graph residual-permutation reference. Declared confidence therefore does not identify the edges driving the harmonic concentration.

The external human-score order provides a component-wise alignment contrast. Among 287 non-tied observed edges, the full elicited logits have log loss 0.4804. Removing h lowers it to 0.4666, an improvement of 0.0138. The admissible null has positive mean improvement 0.0041; the observed gain is 3.4 times that baseline (raw $p = 0.0040$). Removing local curl improves log loss by 0.0052, below the admissible-null mean 0.0081 ($p_{\text{lower}} = 0.286$; $p_{\text{upper}} = 0.714$). The observed curl has a smaller alignment penalty than the mean random admissible curl.

Removing curl still lowers the observed loss; after harmonic flow has already been removed, deleting curl lowers loss a further 0.0057, from 0.4666 to the gradient-only value 0.4609. The observed distinction is comparative: harmonic flow accounts for more of the divergence from the external scalar order than local curl. The harmonic-over-curl specificity contrast is 0.0087 (raw $p = 0.022$).

Projecting directly to the gradient removes both components and gives the largest in-sample improvement, 0.0196, but that gain is typical under the admissible null ($p = 0.122$). This comparison does not distinguish standard HodgeRank scalarization. The harmonic-removal contrast provides a specific hypothesis for independent out-of-sample evaluation.

Brier loss yields the same ordering. Removing h improves it by 0.0054 ($p = 0.0100$), compared with 0.0025 after removing curl ($p = 0.661$); the difference is 0.0029 ($p = 0.043$). This secondary loss points in the same global-versus-local direction.

The probability-scale changes do not materially alter binary decisions: deharmonizing and decurling each change exactly the same four directions (1.39%), and their accuracy specificity contrast is zero.

The edgewise concentration has the same direction as the loss contrast. The LLM direction disagrees with the human-score direction on 75 edges, where mean harmonic energy is 2.171 times its value on agreements, versus null mean 1.165 (raw $p = 0.0090$). In this external scalar comparison, mean harmonic energy is higher on human–LLM directional disagreements than on agreements. The design cannot distinguish transient instability from a stable alternative rubric that differs from the human score.

Taken together with $\rho_\infty = 0$, the component comparisons yield the following pattern: the declarations have zero triangle compatibility radius, local-curl removal produces an alignment improvement within its permutation reference, and global unfilled cycles show the largest unadjusted alignment contrast. Local triangle compatibility and non-scalar global structure coexist in this graph.

Multiplicity limits the evidential strength. The smallest raw tail probabilities in the 17-test family are 0.0040 for deharmonized log loss and 0.0090 for mismatch concentration, followed by 0.0100 for Brier loss. The remaining raw tail probabilities range from approximately .016 to .043. None of the 17 comparisons crosses the familywise reference 0.0029.

Table 8: Confidence-weighted interpretation audit. The declared confidence is an internal diagnostic, not independent ground truth. Null ranges and tail probabilities use the same 2,000 residual permutations accepted as triangle-admissible at the 10^{-9} feasibility tolerance. Tail probabilities are raw Monte-Carlo values; none of the 17 Hodge-related comparisons crosses the .0029 familywise reference.

Statistic	Observed	Null mean	Null 95% range	Relevant tail p
Weighted density ratio T_W	1.430	0.943	[0.621, 1.351]	0.0160
Spearman $\rho(h_e^2, w_e)$	-0.095	-0.128	[-0.230, -0.021]	0.739
Low/high-confidence energy ratio	1.646	1.613	[0.864, 2.770]	0.420

Table 9: External-alignment component ablation. Human scores define an external scalar ordering on 287 non-tied observed edges, not ground truth. Improvements are full-flow loss minus ablated-flow loss. The specificity contrast is the deharmonizing improvement minus the decurling improvement. Tail probabilities are raw Monte-Carlo values; none of the 17 comparisons crosses the .0029 familywise reference.

Statistic	Observed	Null mean	Null 95% range	Upper-tail p
<i>Probability-scale alignment</i>				
Log loss: remove harmonic	0.0138	0.0041	[-0.0024, 0.0108]	0.0040
Log loss: remove curl	0.0052	0.0081	[-0.0021, 0.0179]	0.7141
Log loss: harmonic – curl	0.0087	-0.0040	[-0.0161, 0.0082]	0.0220
Log loss: gradient only	0.0196	0.0123	[-0.0002, 0.0242]	0.1219
Brier: remove harmonic	0.0054	0.0016	[-0.0014, 0.0047]	0.0100
Brier: remove curl	0.0025	0.0035	[-0.0010, 0.0080]	0.6612
Brier: harmonic – curl	0.0029	-0.0019	[-0.0071, 0.0035]	0.0435
<i>Directional decision</i>				
Accuracy: remove harmonic	1.39%	0.27%	[-2.09%, 2.44%]	0.207
Accuracy: remove curl	1.39%	0.78%	[-1.74%, 3.14%]	0.379
Accuracy: harmonic – curl	0.00%	-0.51%	[-3.83%, 2.79%]	0.427
Mismatch/agreement harmonic energy	2.171	1.165	[0.635, 1.931]	0.0090

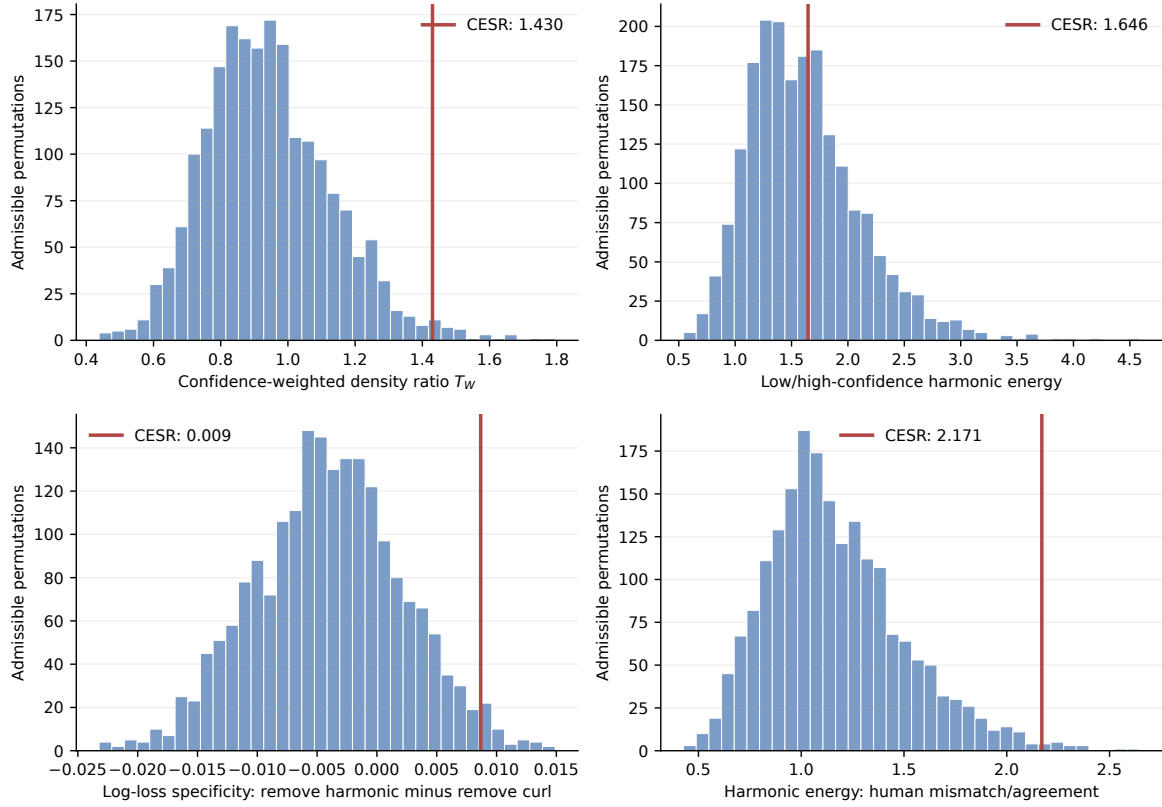


Figure 4: Interpretation audits under 2,000 residual permutations accepted as triangle-admissible at the 10^{-9} feasibility tolerance. Confidence weighting leaves the density ratio nearly unchanged, whereas the low/high-confidence contrast lies within its null range. The lower-left panel compares the alignment gains from removing harmonic flow and local curl; the lower-right panel compares harmonic energy on human-LLM directional disagreements and agreements.

9.5 Tolerance, binary orders, and expected ranks

We report $\varepsilon = .02$ as a sensitivity point and emphasize $\varepsilon = .05$ and $.10$, which match the numerical scale of the measured low-dimensional approximation errors. These tolerances are deterministic robustness scenarios, not estimates of judgment uncertainty or statistical confidence bands.

At $\varepsilon = .05$, the mean expected-rank interval width is 39.05 positions, 66.2% of the complete 59-position range. At $\varepsilon = .10$, it is 43.64 positions, or 74.0%. Coordinatewise rank identification is therefore weak across the displayed robustness range; at $.05$, only 3 of 60 rank projections are narrower than 30 positions. The intervals are projections: their Cartesian product is not jointly attainable.

Rank indeterminacy does not imply that every binary statement is unresolved. At $\varepsilon = .05$, 449 of 1,770 pair orders (25.4%) have a sharp probability interval excluding $.5$. Of these, 154 concern pairs that were not observed directly (10.5% of the 1,466 missing edges). [Table 2](#) reports the rank-width tolerance path and the binary count at $.05$. This is the operational distinction between recovering local order statements and claiming a precise complete ranking.

9.6 A sparsity baseline for rank-interval width

To benchmark rank width at the same object and edge counts, we generate exact Bradley–Terry probabilities for $n = 60$ from $\theta_i = 1.6 - 3.2(i - 1)/(n - 1) + \eta_i$ with independent $\eta_i \sim N(0, .15^2)$, sample 304 unordered edges uniformly subject only to connectivity, and compute zero-tolerance rank projections. Across 30 simulated graphs, the mean width is 39.47 positions (2.5–97.5% empirical range 39.05–39.81), or 66.9% of the available rank range; see [Table 4](#). The corresponding CESR zero-tolerance width is 33.28, which is 6.19 positions narrower. This contrast benchmarks the CESR projections relative to an equal-size random-connected exact-BT reference, but it does not isolate the contribution of edge values because the graph topology and acquisition process differ. In both cases, the remaining uncertainty spans tens of positions.

9.7 Resolution under neighboring assumptions

[Table 10](#) quantifies the comparison while keeping the targets separate. The Crippa–Fedchenko population-SST set is empty on the CESR soft probabilities. The obstruction is already directional: among 299 filled observed triples there is 1 majority cycle, with winning probabilities 0.62, 0.65, 0.66. Its weakest edge is 0.12 above one half, so every common band below $\varepsilon = 0.12$ preserves the cycle and leaves the SST set empty. In particular, neither $.05$ nor $.10$ repairs it. The 30 exact-BT controls have no majority cycle and belong to the SST model by construction. This differs from the triangle result because a distribution over complete rankings may have cyclic majority marginals, whereas the SST model maintains one latent weak order. The directional obstruction rests on a single triple: its weakest edge is $.62$, and the necessary band $.12$ lies $.02$ above the largest displayed tolerance. Repeated, order-balanced measurements of these three edges are therefore needed to assess the stability of the cycle.

The Morel-Balbi and Kirkley MAP fit returns 3 ordered blocks of sizes 20–8–32. Converted to deterministic position spans, its mean is 23.80 positions (40.3% of the amplitude), and 1056/1770 relations cross block boundaries (59.7%). The triangle set at $\varepsilon = .05$ instead has mean expected-rank width 39.05 (66.2%) and determines 449/1770 sharp probability orders (25.4%). On the random connected exact-BT controls, the MAP method still returns only 2.90 blocks on average, with mean span 21.32 (36.1%) and 63.9% cross-block relations; the triangle-set mean expected-rank width is 39.47 (66.9%) at zero tolerance. Although every control is generated by an exact BT model with no ties, the procedure returns two blocks in 3 trials and three in 27. The selected partition has positive log posterior odds against the BT state in 30/30 trials; the odds range from 17.7 to 37.2, with mean 30.1. Block structure under this design therefore cannot by itself be

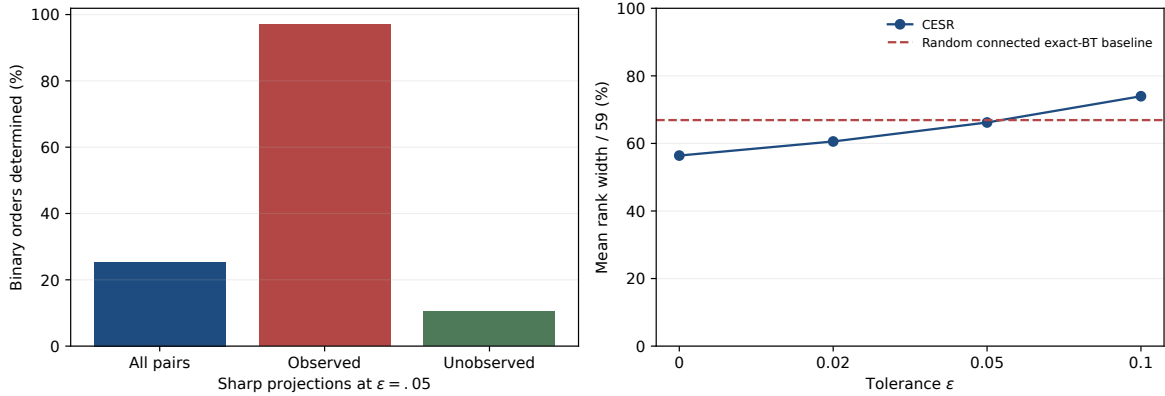


Figure 5: CESR partial identification. Left: fractions of sharp binary-order projections that exclude .5 at $\varepsilon = .05$. Right: mean coordinatewise expected-rank width across tolerances as a fraction of the 59-position amplitude. The dashed line is the separate zero-tolerance random connected exact-BT baseline, not a confidence threshold.

interpreted as evidence of tied latent ranks. No additional tail probability is assigned to these controls.

The equal-size random connected controls also place the MAP summary in context. On the oracle-direction inputs generated from exact-BT probabilities, the MAP span is 36.1%, compared with 40.3% on categorical CESR data; coarse blocks therefore occur even without latent ties.

BT obtains a point order by imposing a known one-dimensional link. The MAP method obtains three coarse blocks by combining a scalar likelihood with a partition prior and discarding the eight tied outcomes. SST would yield a sharp latent-order set under weaker link specification, but the single-declaration CESR data do not satisfy its directional condition. The triangle model accepts the cycle because its target is a random-ranking marginal. At $\varepsilon = .05$, it resolves fewer binary relations than the MAP partition and leaves broad expected-rank intervals. The MAP partition and triangle projections summarize different estimands and forms of uncertainty; their spans should therefore be interpreted within their respective assumptions rather than as estimates of a common resolution target. Posterior uncertainty over partitions is not represented by the reported MAP output.

9.8 Predictive recovery as replication increases

The finite-score three-component PL positive control uses $n = 9$, 27 observed edges out of 36, and 30 simulated truths with fixed seeds. Both BT and mixtures with $K \in \{1, 2, 3, 4\}$ are selected on the complete factorial penalty grid $\gamma \in \{.01, .1, 1\}$ using two disjoint inner validation sets; the selected model is refitted from 12 deterministic starts. We vary independent Bernoulli judgments per observed edge over $\{5, 20, 50, 200\}$ and include exact soft probabilities as the oracle endpoint. Only the nine unobserved true pair probabilities are scored.

At 20 judgments per edge the MSPE contrast is $+0.00034$ $[-0.00094, +0.00199]$. At 50 judgments the point estimate favors the mixture but its interval still includes zero (-0.00141 $[-0.00308, +0.00012]$). At 200 judgments the interval excludes zero (-0.00322 $[-0.00428, -0.00220]$), approaching the oracle contrast -0.00613 $[-0.00738, -0.00494]$. The directional-accuracy interval first excludes zero at 50 judgments. Thus, under this simulation design, a probability-loss improvement is detected at 200 judgments but not at 50.

This simulation illustrates how limited replication can obscure finite-mixture heterogeneity; it does not establish a universal sampling threshold for CESR.

Table 10: Assumption-ladder comparison on CESR and on random connected exact-BT controls matched in object and edge counts. Position span denotes an expected-rank interval width for the triangle set and a deterministic within-block span for the MAP partial ranking. Resolved pairs denote probability intervals excluding .5 and cross-block relations, respectively. The MAP rows report posterior modes, not posterior uncertainty. Dashes denote quantities outside the reported comparison, not zero uncertainty.

Data and method	Output	Mean span (fraction of 59)	Resolved pairs
CESR: BT	Point score order; uncertainty not represented	n/a	n/a
CESR: Morel-Balbi and Kirkley	3 blocks (20–8–32)	23.80 (40.3%)	1056/1770 (59.7%)
CESR: triangle set, $\varepsilon = 0$	Nonempty expected-rank set	33.28 (56.4%)	–
CESR: triangle set, $\varepsilon = .05$	Nonempty expected-rank set	39.05 (66.2%)	449/1770 (25.4%)
CESR: Crippa–Fedchenko SST	Empty: one majority cycle	–	–
Exact-BT controls, oracle-direction input: Morel-Balbi and Kirkley	2.90 blocks; mean log posterior odds 30.1 (range 17.7–37.2)	21.32 (36.1%)	63.9%
Exact-BT controls: triangle set, $\varepsilon = 0$	Nonempty expected-rank set	39.47 (66.9%)	–
Exact-BT controls: Crippa–Fedchenko SST	Nonempty by construction	–	–

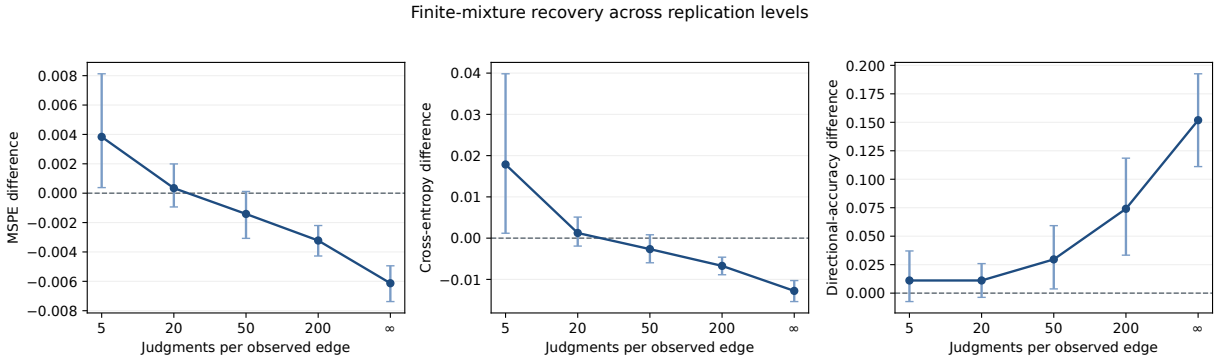


Figure 6: Finite-PL-mixture minus BT contrasts under a correctly specified finite-mixture truth. Negative probability-loss contrasts and positive directional-accuracy contrasts favor the mixture. Whiskers are paired bootstrap intervals over 30 trials.

9.9 Rank-set acquisition

The acquisition experiment tests the proposed identified set directly. At $n = 12$, exact pair marginals from a three-order cyclic mixture are revealed from an initial 11-edge Hamiltonian path in batches of five. Each of 30 paired trials compares Equation (20) with an independently seeded random trajectory. A separate coverage experiment yields mean widths of 8.36 positions at 11 edges, 3.96 at 24, .88 at 48, and zero under full coverage. In the paired acquisition experiment, both trajectories begin at 8.58 positions. The active rule is worse at budgets 16 and 21, then contracts faster from budget 26 onward. At 41 observed edges its mean width is .22 versus 1.36 under random acquisition. It reaches zero mean width at 56 edges; the random mean reaches zero only at full coverage.

The first two batches show that the heuristic does not uniformly dominate random acquisition. Its later gains support using coordinate projections after an initial connected comparison graph has been established.

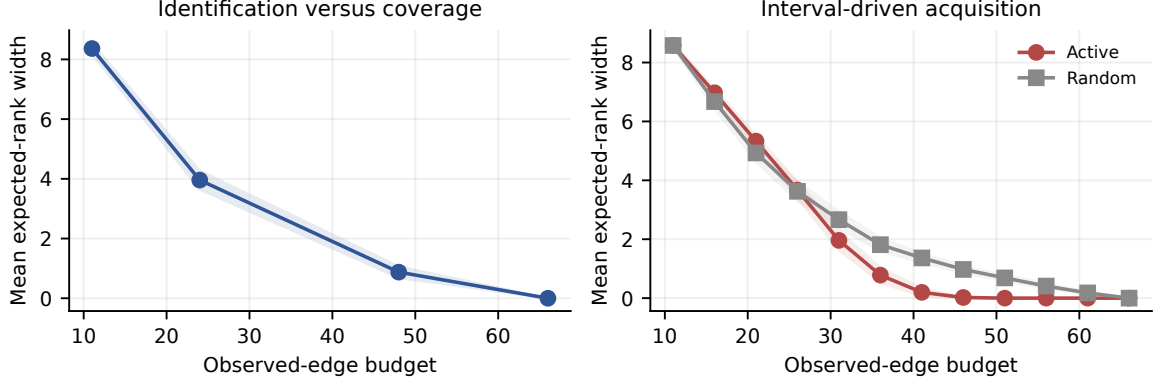


Figure 7: Expected-rank identification versus coverage (left) and interval-driven acquisition (right). Exact synthetic probabilities isolate geometry from judgment noise; bands are pointwise 95% normal-approximation intervals for the trial means.

9.10 Predictive propagation benchmark

The full propagation benchmark is reported in Appendix D. The CESR contrasts, with 2.5–97.5% paired-bootstrap intervals, are

$$\begin{aligned} \text{MSPE} : & \quad -0.000113 \, [-0.000319, +0.000077], \\ \text{cross-entropy} : & \quad -0.000753 \, [-0.001210, -0.000321], \\ \text{directional accuracy} : & \quad +0.003142 \, [+0.000521, +0.006283]. \end{aligned}$$

The accuracy contrast corresponds to approximately .2 of the 64 held-out edges per split. Although the intervals for cross-entropy and directional accuracy exclude zero, the validation sets overlap and share the same LLM target. The effects are therefore practically negligible and do not establish a predictive advantage.

Model selection is equally indeterminate. The modal mixture choice $(K, \gamma) = (2, .01)$ appears in 46.7% of splits, the two inner partitions agree in 30%, and the median candidate gap is only .00033, the same order as the reported loss contrasts. External human-score associations do not discriminate the models either: Spearman coefficients are .768, .768, and .771 and mean rank errors are approximately 8.7 for BT, constrained interpolation, and the finite mixture.

10 Discussion

The paper develops an integrated geometry-to-identification framework for sparse pairwise judgments. First-moment P – Q coupling is feasible for every complementary Q and therefore cannot by itself impose pairwise order geometry. Such restrictions must be placed directly on Q , or through a joint ranking model; in the proposed workflow, P is reconstructed only after Q has been audited and bounded. The three-object coupling counterexample and seven-object rational certificate make the two non-equivalences explicit.

The CESR application illustrates four empirical properties that a point ranking would conflate:

1. the declarations admit a zero-radius completion in the triangle relaxation;
2. neither tested low-dimensional family closely reproduces the observed edge probabilities;
3. the logit flow contains non-gradient structure with distinct local and harmonic components; and
4. most expected ranks have broad identified intervals under the reported sensitivity bands.

The Hodge decomposition characterizes residual topology under the maintained graph geometry.

Harmonic energy is denser per dimension than local-curl energy and has a higher mean on edges whose LLM and human-score directions disagree. Removing the harmonic component yields a larger in-sample loss reduction than removing local curl. None of the 17 Hodge-related comparisons, however, meets the Bonferroni reference 0.0029. These findings motivate independent validation but do not establish a corrective procedure or a behavioral source for the observed structure.

The repeated-judgment experiment shows why a richer parameterization does not by itself make heterogeneous structure predictively learnable. Under the reported simulation design, a probability-loss improvement is detected at 200 judgments per edge but not at 50. This is a design-specific illustration, not a general sampling threshold.

10.1 Scope and limitations

Triangle inequalities characterize the exact linear-ordering polytope only for $n \leq 5$ and form an outer relaxation from $n = 6$ onward. The reported $n = 7$ witness has zero triangle violation and exact L_∞ distance $1/28$ from the convex hull of all 5,040 total orders. Because strictness begins at $n = 6$, this certificate is not dimension-minimal. Consequently, every reported projection is sharp for $\mathcal{T}_n$, not necessarily for $\mathcal{L}_n$. Adding selected higher-order facets can narrow the projections but does not recover the exact linear-ordering polytope unless separation is complete.

The tolerances are deterministic robustness parameters, not confidence intervals. Values .05 and .10 were chosen to match the scale of the low-dimensional approximation errors; those errors do not calibrate judgment uncertainty. The mutable model alias, single call per edge, and absence of A/B counterbalancing also limit interpretation. Estimating prompt variability and order effects requires repeated, counterbalanced judgments from an immutable model snapshot.

Maximum entropy selects one P among all bistochastic matrices with the chosen expected ranks. It is a conditional entropy-maximizing summary, not a posterior over rank positions. The acquisition rule is myopic; its early underperformance in the simulation supports combining it with initial graph exploration.

The curl-harmonic split depends on the observed clique complex, and connected edge-deletion subsampling cannot remove that design dependence. The confidence-weighted sensitivity uses a score declared by the same LLM and is therefore not independent validation. The human-score analysis evaluates alignment with one external scalar criterion, not correctness. The stochastic quantized-BT null preserves the graph and empirical probability support but does not reproduce prompt-order or between-call variability. The residual-permutation null additionally assumes edge exchangeability, while conditioning accepted draws on numerical triangle admissibility at the 10^{-9} feasibility tolerance. None of the 17 Monte-Carlo comparisons meets the Bonferroni reference; independent data and a contrast specified in advance are needed for confirmation.

The comparison with neighboring methods is assumption-indexed because the methods target different estimands. The SST obstruction depends on one observed majority cycle, and the MAP procedure reports a single posterior mode rather than posterior uncertainty. Repeated, order-balanced measurements would be needed to assess the stability of the cycle and the broader assumption-ladder comparison.

10.2 Assessment use

Pairwise LLM assessment can affect consequential decisions. The workflow is a decision-support audit, not an autonomous grading system. Operational use requires model-identified repeated judgments, prompt-order tests, subgroup fairness analysis, data governance, human review, and an appeal path. Wide rank projections should trigger more evidence rather than arbitrary tie-breaking.

11 Conclusion

Sparse pairwise judgments need not support the precision of a complete ranking. The proposed framework makes that statement computable: audit triangle compatibility in Q under the declared relaxation, project the compatible set onto pair probabilities and expected ranks, reconstruct P only conditionally, and use unresolved projections to guide further comparisons.

In the CESR application, the 304 elicited probabilities admit a zero-radius completion in the triangle relaxation but remain approximately .05 in mean absolute error from both tested low-dimensional representations. Sensitivity analyses at $\varepsilon = .05$ and .10 yield broad expected-rank projections, although some binary relations remain identified. At zero tolerance, the CESR mean expected-rank interval width is 33.28 positions, compared with 39.47 on random connected exact-BT graphs matched in object and edge counts. Both sets of projections remain broad; this comparison does not isolate the contribution of edge values because graph topology differs. A MAP tie-block procedure also returns blocks on oracle-direction inputs from the no-tie exact-BT controls, underscoring the distinct assumptions behind point, partition, and set-valued summaries.

The Hodge decomposition detects non-gradient structure, while the component comparisons do not survive familywise multiplicity correction. The controlled repetition experiment supports a circumscribed conclusion: under that finite-mixture design, a probability-loss improvement is detected at 200 judgments per edge but not at 50. Together, these findings show that local triangle compatibility, low-dimensional representability, and rank identification are distinct empirical properties.

A Additional proofs and derivations

A.1 Permutahedron characterization

For completeness, sort any vector $x \in \mathbb{R}^n$ increasingly as $x_{(1)} \leq \dots \leq x_{(n)}$. A standard majorization characterization gives

$$x \in \Pi_n \iff \sum_{i=1}^k x_{(i)} \geq \frac{k(k+1)}{2} \quad (k = 1, \dots, n-1), \quad \sum_{i=1}^n x_{(i)} = \frac{n(n+1)}{2}. \quad (23)$$

The subset inequalities in the proof of [Theorem 3](#) imply these sorted partial sums by selecting the indices of the k smallest coordinates. Conversely, the sorted inequalities imply every subset inequality. This makes the argument independent of a chosen label order.

A.2 Why the expected-rank link is not joint realizability

The marginal map

$$\mu \mapsto (P(\mu), Q(\mu)) \quad (24)$$

has an image equal to the convex hull of paired vertices (M^π, X^π) . Requiring $P \in \mathcal{B}_n$, $Q \in \mathcal{L}_n$, and matching first moments replaces that joint polytope by an intersection of separate marginal polytopes with only n linking equations. In general, separate marginal feasibility does not imply the existence of a coupling supported on compatible pairs. The gap is analogous to the distinction between locally consistent marginals and a global joint distribution in graphical models.

An explicit three-object counterexample makes the failure concrete. Let

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 3/4 & 0 & 1/4 \\ 1/4 & 0 & 3/4 \end{pmatrix}, \quad q_{12} = q_{13} = 1/2, \quad q_{23} = 1. \quad (25)$$

Both marginals yield expected ranks $(2, 1.5, 2.5)$. Yet P places object 1 in position 2 with probability one, while $q_{23} = 1$ requires object 2 to precede object 3 almost surely. Every compatible ranking would then be $2 \prec 1 \prec 3$, which forces $q_{12} = 0$, contradicting $q_{12} = 1/2$. Thus the expected-rank link is insufficient even at $n = 3$.

A.3 An exact relaxation-gap certificate

For $n = 7$, orient coordinates lexicographically as

$$(12, 13, 14, 15, 16, 17, 23, 24, 25, 26, 27, \\ 34, 35, 36, 37, 45, 46, 47, 56, 57, 67).$$

Let x^* be the vector

$$(1, 1, 1, \frac{1}{2}, 1, 1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, 0, 0, 0, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}) \quad (26)$$

satisfying every box and triangle inequality. The following rational primal–dual certificate proves, without relying on floating-point LP output, that its exact L_∞ distance from $\mathcal{L}_n$ is $1/28$. Define

$$\ell(x) = -x_{15} + x_{16} + x_{17} - x_{23} + x_{25} + x_{26} - x_{27} + x_{34} - x_{35} - x_{36} - x_{37} + x_{45} - x_{46} + x_{47}.$$

Direct enumeration of all $7! = 5040$ total-order vertices v gives $\ell(v) \leq 3$, while the displayed witness x^* has $\ell(x^*) = 7/2$. Since the coefficient ℓ_1 norm is 14, every $y \in \mathcal{L}_n$ satisfies $\|y - x^*\|_\infty \geq (7/2 - 3)/14 = 1/28$.

For the matching upper bound, let $v(\pi)$ denote the incidence vector of the listed order π . The following multiplicities sum to 28:

$$\begin{aligned} &1(1264753) + 1(1472563) + 3(1645732) + 3(1647253) + 1(1726345) \\ &+ 3(1726453) + 3(1726534) + 1(5163472) + 3(5164732) + 1(5167342) \\ &+ 2(5172634) + 1(5173264) + 3(5176324) + 1(6451732) + 1(7251634). \end{aligned}$$

The rational convex combination $y = 28^{-1} \sum_{\pi} m_{\pi} v(\pi)$ satisfies $\|y - x^*\|_\infty = 1/28$. The lower and upper bounds coincide. The archive regenerates the 5,040 vertex checks, the 15 support rows, and every coordinate of this rational combination.

B Linear-programming specification

Store one variable $x_{ij} = q_{ij}$ for each $i < j$. Reverse orientations are recovered by $q_{ji} = 1 - x_{ij}$. Every constraint can therefore be written $Ax \leq b$.

For each triple $i < j < k$, Equation (11) gives

$$-x_{ij} - x_{jk} + x_{ik} \leq 0, \quad (27)$$

$$x_{ij} + x_{jk} - x_{ik} \leq 1. \quad (28)$$

An observed-edge tolerance contributes

$$x_{ij} \leq \hat{p}_{ij} + \varepsilon_{ij}, \quad (29)$$

$$-x_{ij} \leq -\hat{p}_{ij} + \varepsilon_{ij}. \quad (30)$$

For reversed observations, replace the target by $1 - \hat{p}_{ji}$. Variable bounds are $0 \leq x_{ij} \leq 1$.

The rank objective for item i is

$$r_i(x) = 1 + \sum_{j < i} x_{ji} + \sum_{j > i} (1 - x_{ij}), \quad (31)$$

under the stored orientation $x_{ab} = q_{ab}$ for $a < b$. Constants can be dropped from the LP objective. The compatibility-radius LP appends $t \geq 0$ and replaces each edge tolerance with $|x_{ij} - \hat{p}_{ij}| \leq t$.

C Algorithms

Algorithm 1: Triangle compatibility audit and rank identification

1. Orient every observed comparison as $i < j$ and construct the sparse triangle matrix.
2. Solve Equation (13). If ρ_∞ exceeds the audit tolerance, report the minimal correction and inspect high-residual edges.
3. Choose edge tolerances ε_{ij} and form $\mathcal{Q}_\varepsilon$.
4. For each object i , minimize and maximize $r_i(Q)$ over $\mathcal{Q}_\varepsilon$.
5. Optionally bound every unobserved q_{ij} for acquisition or sensitivity analysis.
6. Report intervals before any point ranking.

Algorithm 2: Point completion and conditional P reconstruction

1. Fit centered Bradley–Terry scores on observed soft judgments. Use the documented positive ridge for the displayed anchor; an unpenalized anchor is permitted only on a connected comparison graph.
2. Construct Q^{BT} for all pairs.
3. Solve Equation (19) over $\mathcal{T}_n \cap \mathcal{Q}_\varepsilon$ to obtain $\hat{Q}$.
4. Compute $\hat{r}_i = 1 + \sum_{j \neq i} \hat{q}_{ji}$.
5. Solve Equation (17) to obtain a maximum-entropy summary $\hat{P}$.
6. Keep $\hat{P}$ conceptually separate from the LP rank intervals.

Algorithm 3: Interval-driven acquisition

1. Compute current rank widths d_i from Algorithm 1.
2. For every candidate pair, compute pair bounds $[\underline{q}_{ij}, \bar{q}_{ij}]$.
3. Rank candidates by $A(i, j)$ in Equation (20).
4. Acquire the top batch, update E , and repeat.
5. Use randomized exploration at small budgets because uniform dominance is not guaranteed.

D Held-out predictive-propagation benchmark

The held-out benchmark asks whether a finite Plackett–Luce mixture predicts unobserved pair probabilities better than Bradley–Terry. It is not used by the triangle compatibility radius, identified set, rank projections, acquisition rule, or conditional P reconstruction. A K -component model has pair marginal

$$q_{ij}^{\text{mix}} = \sum_{h=1}^K \omega_h \text{sigmoid}(\theta_{hi} - \theta_{hj}), \quad \omega_h \geq 0, \quad \sum_h \omega_h = 1,$$

and $Kn - 1$ free parameters after centering component scores and normalizing weights. Synthetic experiments use the complete factorial grid $K \in \{1, 2, 3, 4\}$ and $\gamma \in \{.01, .1, 1\}$. CESR uses the common grid $\gamma \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ for both mixture and BT. Selection minimizes mean cross-entropy over two disjoint inner edge sets whose complementary training graphs are connected; MSPE is the deterministic tie-breaker. Selected fits use 12 deterministic starts, and the archive retains every objective, candidate gap, split winner, and held-out prediction.

The synthetic battery contains a scalar-to-three-order cyclic continuum, nine density–topology cells at each endpoint, two deliberate misspecifications, and the finite-score repetition study reported in the main text. Only unobserved edges are used for predictive claims. The following figures report predictive behavior while the main text remains centered on geometry and identification.

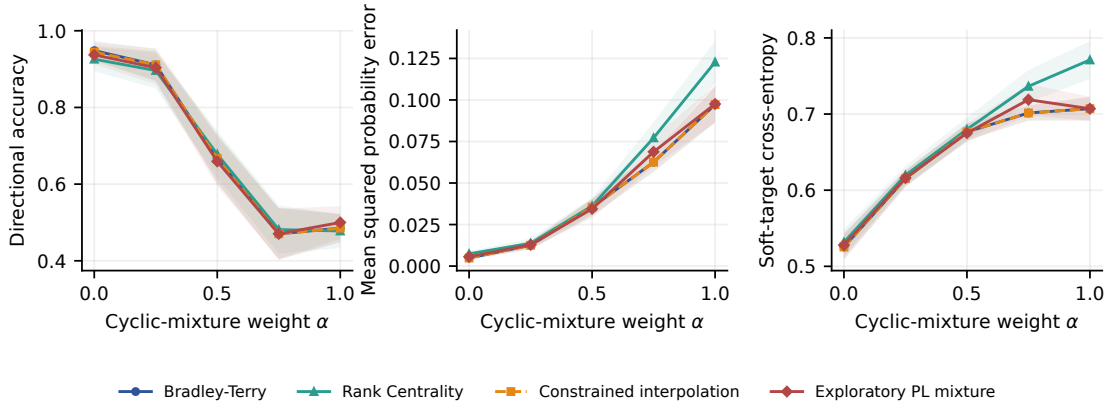


Figure 8: Strictly held-out prediction along the scalar-to-cyclic continuum. No probability-loss contrast at the cyclic endpoint excludes zero.

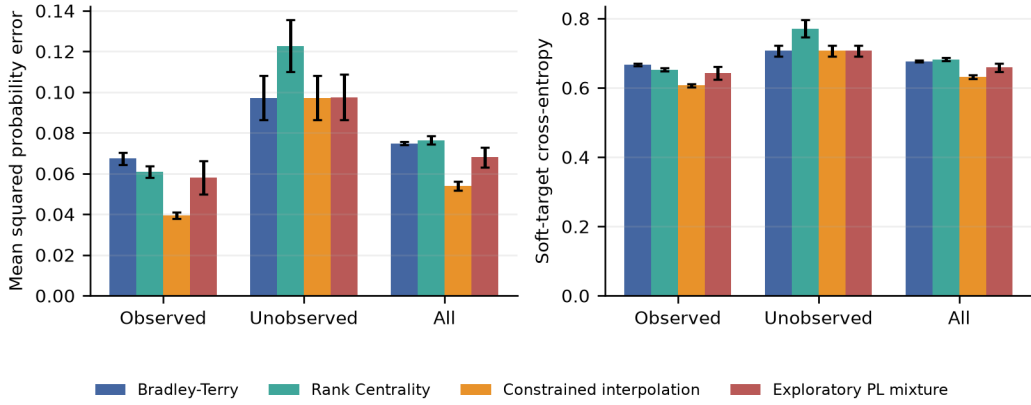


Figure 9: Observed, unobserved, and all-pair decomposition. The apparent all-pair gain of constrained interpolation is observed-edge fit rather than predictive propagation.

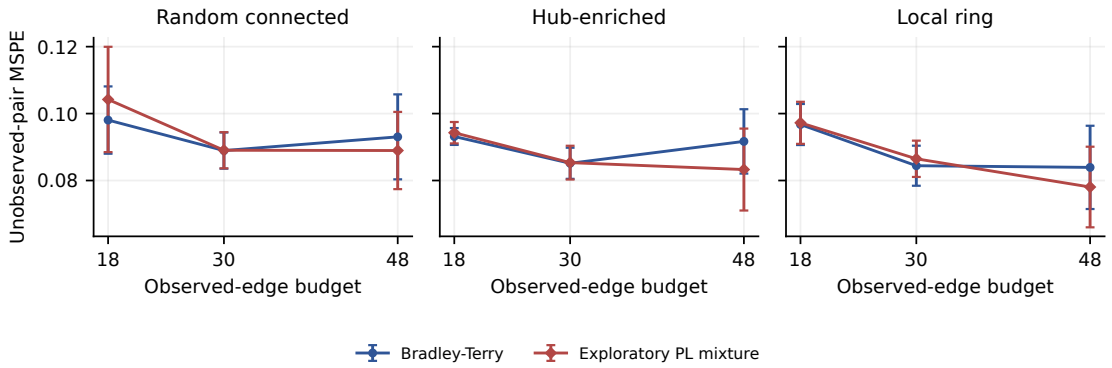


Figure 10: Unobserved-pair MSPE at the cyclic endpoint across the nine density-topology settings. Contrasts change sign across cells and do not reveal a stable graph-structure effect.

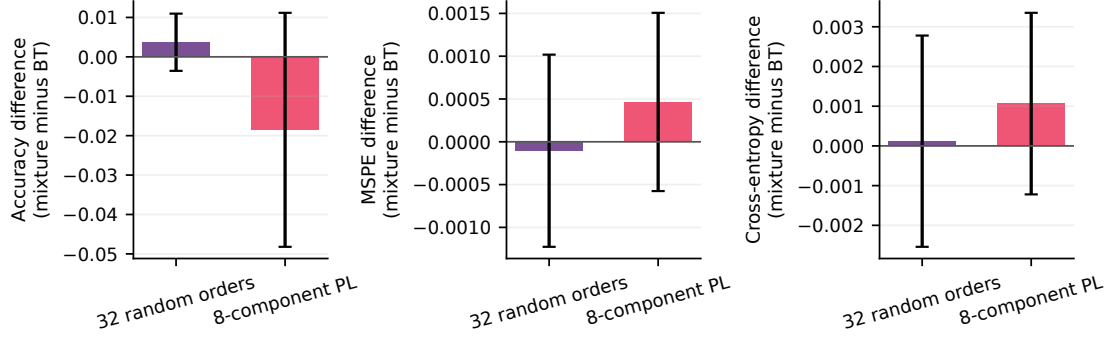


Figure 11: Mixture-minus-BT contrasts under a diffuse 32-order truth and an eight-component finite-PL truth. Neither regime supports a stable advantage.

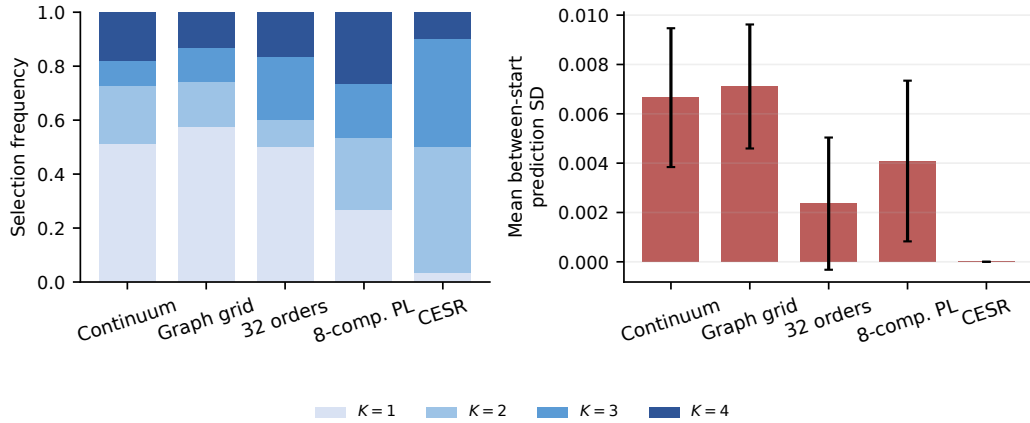


Figure 12: Selected component counts and between-restart prediction dispersion. Predictive similarity does not identify latent components.

D.1 Detailed numerical results and diagnostics

All contrast signs are left method minus right method. Negative MSPE and cross-entropy contrasts favor the left method; positive directional-accuracy contrasts favor it. Bootstrap percentiles use the resampling units stated in the text. The archive contains the trial rows, all held-out predictions, the complete tuning grid, raw restart objectives, edgewise restart dispersion, and full selection frequencies.

Table 11: Paired contrasts at the deterministic-order cyclic endpoint.

Metric	Mean	2.5%	97.5%
Directional accuracy	+0.01481	-0.00370	+0.03704
MSPE	+0.00035	-0.00525	+0.00619
Soft-target cross-entropy	-0.00004	-0.01645	+0.01561

Table 12: All 18 graph-grid MSPE contrasts on unseen pairs. Negative values favor the mixture.

Endpoint	Topology	Edges	Mean	2.5%	97.5%
Scalar	Random connected	18	-0.00137	-0.00236	-0.00053
Scalar	Random connected	30	+0.00114	+0.00025	+0.00218
Scalar	Random connected	48	+0.00042	-0.00001	+0.00098
Scalar	Hub-enriched	18	+0.00106	-0.00041	+0.00316
Scalar	Hub-enriched	30	+0.00063	-0.00059	+0.00226
Scalar	Hub-enriched	48	+0.00031	-0.00036	+0.00107
Scalar	Local ring	18	-0.00028	-0.00259	+0.00226
Scalar	Local ring	30	-0.00032	-0.00081	+0.00021
Scalar	Local ring	48	-0.00056	-0.00114	+0.00001
Cyclic	Random connected	18	+0.00616	+0.00104	+0.01287
Cyclic	Random connected	30	+0.00009	-0.00000	+0.00027
Cyclic	Random connected	48	-0.00408	-0.01243	+0.00373
Cyclic	Hub-enriched	18	+0.00113	+0.00000	+0.00289
Cyclic	Hub-enriched	30	+0.00019	-0.00133	+0.00162
Cyclic	Hub-enriched	48	-0.00841	-0.01633	-0.00157
Cyclic	Local ring	18	+0.00051	-0.00255	+0.00347
Cyclic	Local ring	30	+0.00207	+0.00006	+0.00424
Cyclic	Local ring	48	-0.00587	-0.01437	+0.00074

Across the 18 density–topology cells in [Table 12](#), contrast signs vary across both endpoint families, with nominal intervals excluding zero in opposite directions. No consistent graph-structure pattern emerges. The CESR modal selection (2, .01) occurs in 46.7% of splits, inner-split agreement is 30%, and the median candidate gap is .00033; the data do not determine the selected finite-mixture specification.

Table 13: Misspecification contrasts: selected finite-PL mixture minus BT.

Truth	Metric	Mean	2.5%	97.5%
32 random orders	Directional accuracy	+0.00370	+0.00000	+0.01111
32 random orders	MSPE	-0.00010	-0.00116	+0.00103
32 random orders	Soft-target cross-entropy	+0.00012	-0.00228	+0.00290
Eight-component PL	Directional accuracy	-0.01852	-0.05185	+0.00741
Eight-component PL	MSPE	+0.00047	-0.00047	+0.00159
Eight-component PL	Soft-target cross-entropy	+0.00107	-0.00095	+0.00358

Table 14: Model-selection diagnostics by experimental family. Agreement is the fraction of fits whose two inner partitions select the same complete (K, γ) pair. Gap is the median [IQR] of second-best minus best mean validation cross-entropy. The complete frequency table is machine-readable.

Family	Fits	Modal (K, γ)	Frequency	Agreement / gap
CESR	30	(2,0.01)	46.7%	30.0% / 0.00033 [0.00017,0.00058]
Continuum, $\alpha = 0$	30	(1,0.01)	20.0%	16.7% / 0.00091 [0.00033,0.00265]
Continuum, $\alpha = .25$	30	(1,0.01)	26.7%	3.3% / 0.00084 [0.00027,0.00259]
Continuum, $\alpha = .5$	30	(1,1)	23.3%	16.7% / 0.00094 [0.00056,0.00211]
Continuum, $\alpha = .75$	30	(1,1)	53.3%	23.3% / 0.00716 [0.00276,0.01194]
Continuum, $\alpha = 1$	30	(1,1)	76.7%	30.0% / 0.01274 [0.00571,0.02299]
32 random orders	30	(1,1)	33.3%	26.7% / 0.00050 [0.00013,0.00253]
Eight-component PL	30	(4,1)	16.7%	13.3% / 0.00037 [0.00011,0.00115]
Graph grid, cyclic	135	(1,1)	71.9%	44.4% / 0.01164 [0.00529,0.02163]
Graph grid, scalar	135	(1,0.01)	32.6%	9.6% / 0.00073 [0.00034,0.00172]
Finite PL, binomial	30	(1,1)	36.7%	10.0% / 0.00059 [0.00029,0.00140]
Finite PL, oracle	30	(4,0.01)	66.7%	40.0% / 0.00198 [0.00073,0.00332]

Table 15: Full-data CESR geometric audit of the selected finite-PL mixture.

Quantity	Value
Selected components / penalty	2 / 0.01
Parameters / observed edge	0.391
Mean / maximum observed absolute error	0.0544 / 0.2899
Edges outside $\varepsilon = .02$	219 / 304
Membership in $\mathcal{T}_n$ / $\mathcal{Q}_{.02}$	yes / no
Mean all-pair absolute distance to BT / constrained point	0.0110 / 0.0164
Maximum triangle violation	0.00e+00

Table 16: In-sample association with human scores. Spearman and Kendall correlate $-\hat{r}$ with the 60 human scores; Kendall uses tau-b for ties.

Model	Spearman	Kendall tau-b	Mean rank error
Bradley-Terry	0.768	0.590	8.717
Constrained interpolation	0.768	0.589	8.669
Selected finite-PL mixture	0.771	0.593	8.618

Table 17: CCSR tolerance sensitivity. Mean correction is $|E|^{-1} \sum_{(i,j) \in E} |\hat{q}_{ij} - \hat{p}_{ij}|$.

ε	Mean width	Median	Maximum	Mean correction	Maximum correction
0	33.278	35.335	40.360	0.0000	0.0000
0.005	33.907	35.958	40.910	0.0048	0.0050
0.01	34.532	36.575	41.440	0.0090	0.0100
0.02	35.739	37.700	42.580	0.0158	0.0200
0.05	39.054	40.720	45.470	0.0289	0.0500
0.1	43.638	45.395	49.060	0.0359	0.1000

Table 18: Four-point sensitivity to the constrained-interpolation reference strength on one fixed CCSR split. Validation predictions are invariant to the displayed precision, whereas observed-edge corrections vary with the reference strength.

λ	Validation MSPE	Validation cross-entropy	Mean observed correction
0.1	0.010327	0.484970	0.00074
1	0.010327	0.484970	0.00654
10	0.010327	0.484970	0.01486
100	0.010327	0.484970	0.01649

E Additional geometric checks

E.1 Triangle relaxation versus the exact polytope

The seven-object fractional witness satisfies all triangle inequalities to numerical tolerance but has L_∞ distance $1/28$ from the convex hull of the 5,040 total-order vertices. The archive enumerates those vertices on every check. This confirms that triangle-relaxed intervals are conservative outer-model projections, not exact linear-ordering-polytope projections.

E.2 Tolerance and reference-strength definitions

Table 17 uses mean observed absolute correction,

$$|E|^{-1} \sum_{(i,j) \in E} |\hat{q}_{ij} - \hat{p}_{ij}|.$$

The maximum absolute correction is also shown. At $\varepsilon = .02$, mean, median, and maximum coordinatewise expected-rank widths are 35.739, 37.700, and 42.580 positions. This row is reported as a sensitivity analysis; the main text emphasizes .05 and .10. Small differences from prose values reflect display rounding.

The four-point λ sensitivity in Table 18 uses one fixed connected 240/64 CCSR split. Predictions on inactive unseen coordinates agree with Bradley–Terry to numerical precision for all four values, while the mean observed correction changes. Across all six pairs of strengths in $\{.1, 1, 10, 100\}$, the largest absolute difference on a validation prediction is $4.46\text{e-}08$. Held-out performance therefore does not discriminate among these values; we use $\lambda = 10$ as a fixed reconstruction setting.

Table 19: Candidate-level restart sensitivity across ten simulated instances of four specified settings. Agreement is with the complete (K, γ) choice at two starts. Gap is second-best minus best mean inner-validation cross-entropy.

Setting	Starts	Agreement	Mean gap	Mean Δ MSPE
Finite PL oracle	2	100%	0.00247	-0.00648
	6	90%	0.00204	-0.00668
	12	90%	0.00204	-0.00668
Cyclic endpoint	2	100%	0.01292	+0.00366
	6	80%	0.01486	+0.00503
	12	80%	0.01351	+0.00626
Graph cyclic, random/30	2	100%	0.01281	+0.00014
	6	100%	0.01281	+0.00014
	12	100%	0.01281	+0.00014
CESR	2	100%	0.00026	-0.00049
	6	100%	0.00026	-0.00049
	12	100%	0.00026	-0.00049

Table 20: Secondary CESR penalty-grid sensitivity using an expanded grid that includes $\gamma = .001$. Intervals are paired-bootstrap percentiles over the 30 overlapping outer splits.

Metric	Mean	2.5%	97.5%
MSPE	-0.00014	-0.00035	+0.00004
Cross-entropy	-0.00081	-0.00126	-0.00038

F Reproducibility and numerical safeguards

All stochastic seeds are deterministic and all truth parameters are serialized as JSON Lines. SciPy/HiGHS solves the linear programs; CLARABEL solves exponential-cone programs with SCS fallback. The run manifest records Python and package versions, trial counts, the full factorial grid, validation-edge indices, graph-connectivity flags, validation overlap, restart budgets, and the exact directional-accuracy rule. Feasibility residuals are checked after each constrained optimization and retained in the relevant audit tables.

After creating a Python environment and installing the supplied requirements, the public replication package provides two targets:

```
make public-smoke PYTHON=.venv/bin/python
make public-paper PYTHON=.venv/bin/python
```

The first target reruns the public synthetic and certificate checks; the second compiles the article PDF from the included aggregate tables and figures. Regenerating the CESR analyses requires authorized access to the restricted data. Regenerating the neighboring-method benchmark additionally requires the external implementation identified in the package documentation.

The Python requirements do not install system build tools. Full PDF regeneration additionally requires GNU Make, a POSIX-compatible shell, a TeX distribution, `latexmk`, `pdflatex`, and `bibtex`; the validated Python and package versions are recorded in the execution manifest. The archive’s `DATA_STATEMENT.md`, `DATA_LICENSE.md`, and `LICENSE` distinguish code, synthetic outputs, and pseudonymized CESR derivatives. Free-text LLM justifications, original file names, persistent storage identifiers, source documents, student text, and the re-identification key are not distributed.

The CESR cache contains pseudonymous identifiers, model probabilities, confidence values, and categorical preferences. Student text, free-text justifications, and private exam material are

excluded. The acquisition protocol records the model alias, API endpoint, sampling configuration, truncation, clipping, and absence of pair-order counterbalancing.

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